



**MASTERMIND
INDIA**

Spirit of India Special

SPIN DOCTOR

By Amitabh Srivastava

A dash of grey stubble and restless eyes speak volumes for Madhabanada Ray. With a masters in Agriculture and a PG diploma in Tea Plantation Management, Ray has always been willing to go the extra mile to gain a new skill. This perhaps explains how Madhab, as friends call him, who had never shown any inclination for design, has now created a tasar yarn reeling machine to help rural women produce a better quality of yarn with an easier production process, translating into more money for them.

The new reeling machine produces many types of yarns—both untwisted (used in wefts) and twisted that is used as warp—with a better quality to offer to the market. Not only is this machine compact and better managed, it also operates through power-driven gears, weeding out the pedal system. It can reject thicker yarns, knots and flosses that enhance the quality of yarn. “The machine is better in multiple aspects than the one provided by the Central Silk Board (CSB),” says Ray, MD, MASUTA (Mahila Sutra Tasar) Producers Company Limited that works with the yarn producers.

The user can also work on any number of spindles, stop any of the spindles without stopping the whole machine, thereby increasing the productivity of the machine. The cost at mass production scale would be less than the existing reeling machines. “Necessity is rightly called the mother of all invention,” Ray says. “I was unhappy watching the rural women getting exhausted at the earlier reeling machines that operate on pedals. It was too large and too

cumbersome and had quite a few limitations,” he recalls.

The new solar power-operated reeling machine that he has introduced has multiple advantages in terms of having manageable weight and height with better placement position, improved safety and less power consuming motors. All this guarantees improved productivity and production flexibility.

Like most innovations, this one too did not come easily. Ray had to do everything—from meeting physics and mechanics professors in various cities to chatting on the Internet with students of The Netherlands’ Delft University of Technology to gather technical expertise. Impressed by MASUTA’s goals of economically empowering village women, some students from Delft University visited India to help in the development. “They helped us manufacture the first prototype. Then in the second prototype, we focused on reducing power requirements. In the final stage, we worked towards bettering the bobbin,” he recalls.

Once the model was ready, Ray kept introducing the desired modifications in all the stages to fine-tune it further. “Finally we are ready with the final product,” says Ray. He is hopeful that the enterprise will increase the users’ income by 30 per cent. The earning goes into the hands of the women of the family. Ray’s life changed with a deciding clarity when he joined an NGO, PRADAN (Professional Assistance from Development Action), that works with communities at the grassroots to help them overcome poverty. Ray worked as a rural development professional in 1995. “That’s destiny. I wanted to make a meaningful change in the lives of the



**Madhabanada
RAY, 39**

INNOVATION

**A new tasar yarn
reeling machine**

COST OF PRODUCTION

Rs 30,000

(including the cost of the machine)

WHAT IT DOES

**It operates through
power-driven gears,
replacing the pedal
system and ensuring
higher productivity**

**RAY'S INNOVATION HAS
CHANGED THE LIVES OF
HUNDREDS OF VILLAGE WOMEN**